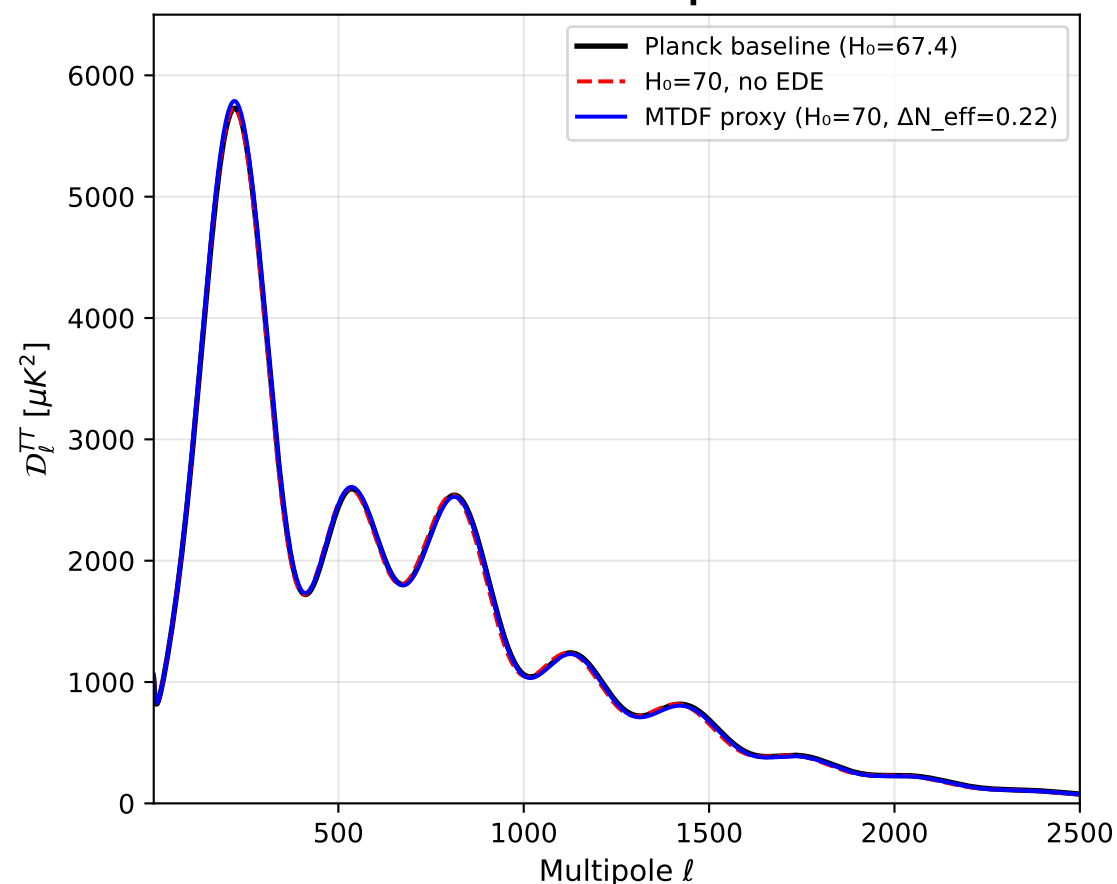
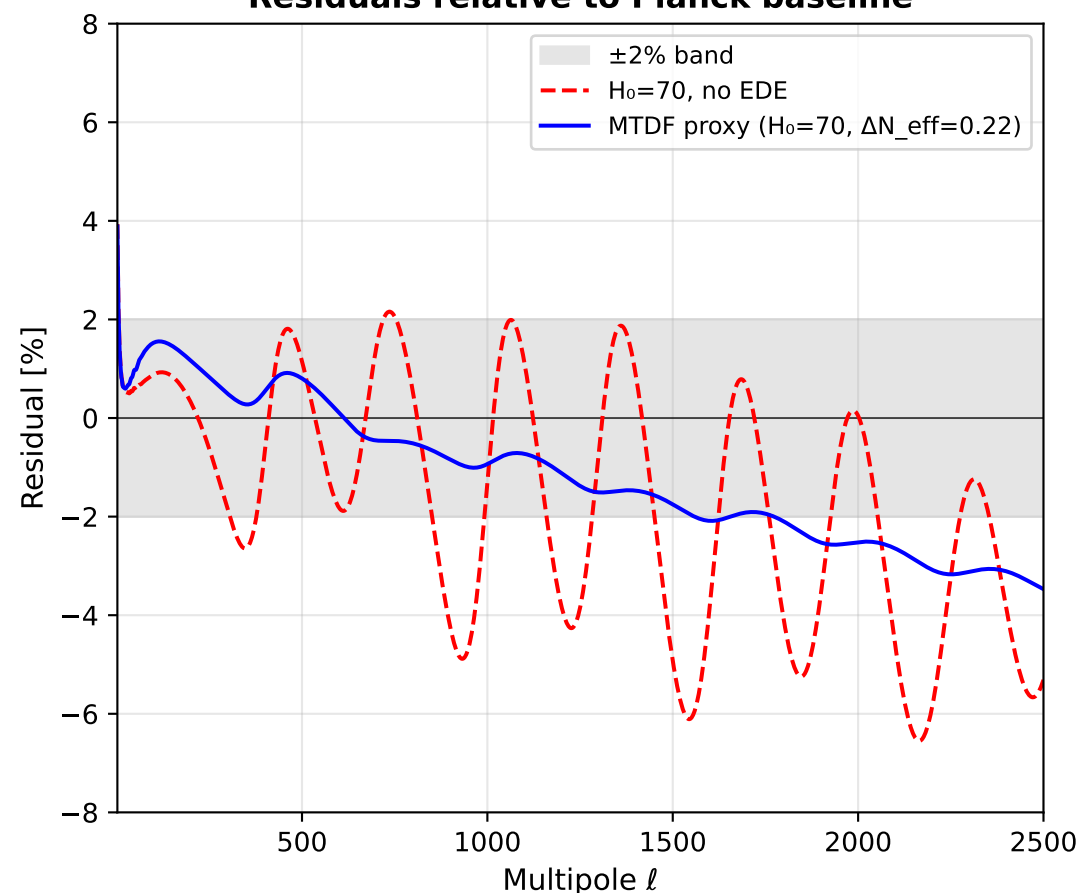


MTDF Proxy in CLASS: CMB TT Analysis

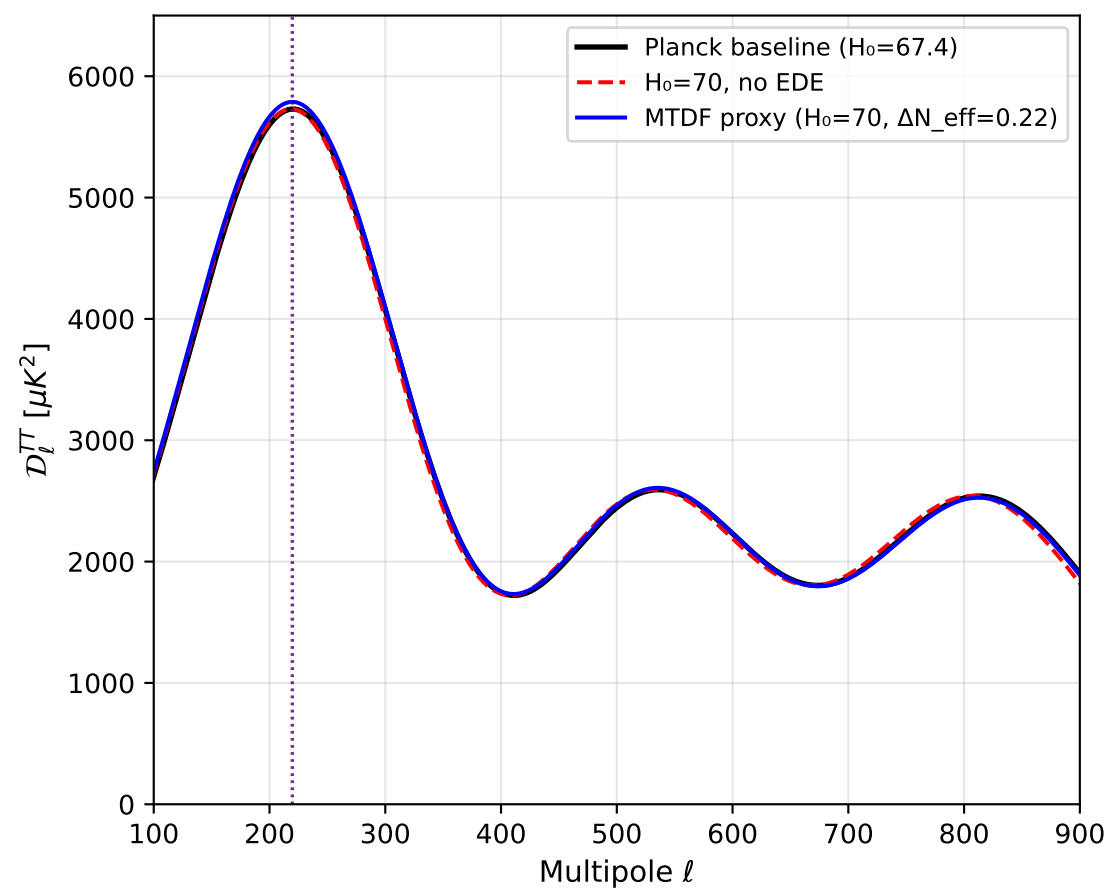
CMB TT Power Spectrum



Residuals relative to Planck baseline



First 3 Acoustic Peaks



MTDF CLASS Analysis Summary

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chi2 computed over $30 \leq l \leq 2000$, 50 bins

Model	H0	dNeff	r_d	100*th_s	chi2_TT	l_peak
Planck baseline	67.36	0.00	147.10	104.1850	0.0	220
$H_0=70$ no EDE	70.00	0.00	147.10	104.9688	78.1	219
MTDF proxy	70.00	0.22	146.00	104.1953	24.2	220

Key findings:

- * r_d reduction (MTDF vs $H_0=70$): -0.74%
- * χ^2 improvement: 78.1 $\rightarrow$ 24.2 (factor 3.2x)
- * θ_s recovered: 104.1953 vs 104.1850 (baseline)
- * First peak restored to $l = 220$

Notes:

- * r_d = sound horizon at drag epoch (used for BAO)
- * $\Delta N_{\text{eff}} = 0.22$ mimics MTDF early field energy
- * χ^2 uses diagonal approx with $\sigma \sim 2\% D_l$